

REVIEW

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Business model innovation in SMEs for sustainable and digital transformation: a systematic review and integrative framework

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Abstract

Small- and medium-sized enterprises (SMEs) increasingly need to reconfigure how they create, deliver, and capture value in response to digitalisation, sustainability demands, and environmental uncertainty. This study systematically reviews 162 English-language articles published in SSCI-indexed journals to integrate fragmented research on business model innovation (BMI) in SMEs. Descriptive analysis, keyword co-occurrence analysis, and article-level qualitative content analysis are combined. The findings identify four broad knowledge domains and show that 41 studies treat BMI primarily as an outcome, 44 as an organisational process, 17 as an antecedent of subsequent outcomes, and 36 as part of a combined causal relationship; a further 24 studies are descriptive or typological. Across these roles, the literature explains BMI through resource mobilisation, entrepreneurial action logics, absorptive and dynamic capabilities, experimentation, and network mobilisation. Its effects on performance, growth, resilience, internationalisation, and sustainability depend on complementary resources, implementation capabilities, stakeholder alignment, and environmental conditions. The evidence supports an adaptive and iterative interpretation of SME BMI, but direct evidence of recursive feedback remains limited. Resource constraints also have conditional effects, stimulating bricolage and focused experimentation in some circumstances while inhibiting substantive change when shortages become severe. The review provides an integrative synthesis specific to SMEs and advances digital business model innovation research by explaining how digital pressures are translated into changes in value proposition, value creation and delivery, and value capture through organisational and relational mechanisms.

Keywords Business model innovation, Digital business model innovation, SMEs, Systematic literature review, Dynamic capabilities, Digital transformation

Introduction

Small- and medium-sized enterprises play an important role in economic development, industrial upgrading, technology diffusion, and employment creation [22, 34, 66]. Their business model decisions are shaped by conditions that differ from those of larger firms, including limited financial and managerial resources, restricted access to specialised technologies, concentrated decision authority, and greater dependence on external partners [38, 70]. These conditions do not produce a uniform innovation response. Organisational flexibility and close customer relationships may facilitate rapid adaptation in some SMEs, whereas severe shortages of finance, skills,

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managerial attention, or external support may restrict their capacity to undertake substantive change [39, 48]. Understanding innovation in SMEs therefore requires attention not only to resource availability but also to how firms mobilise, combine, and supplement constrained resources.

This issue has become increasingly important as digitalisation and sustainability pressures alter the conditions under which firms create and appropriate value. A business model describes the architecture through which an enterprise creates, delivers, and captures value [68]. Business model innovation (BMI), accordingly, refers to purposeful and non-trivial changes to that architecture or to the activity system through which value is produced and appropriated [2, 24]. Within this broader domain, digital business model innovation (DBMI) refers to business model change in which digital technologies are integral to the reconfiguration of value proposition, value creation and delivery, or value capture [3, 77]. Digital technology adoption alone therefore does not constitute DBMI; it becomes relevant to business model innovation when it changes how a firm offers value, organises value creation and delivery, or appropriates returns. For SMEs, BMI may involve redesigning value propositions, adopting digital channels or service models, reorganising relationships with customers and partners, developing circular or sustainability-oriented activities, or changing revenue and value-capture mechanisms [3, 85]. In this review, DBMI is treated as an important digitally enabled form of BMI within the broader SME BMI literature rather than as a separate review domain. These changes may support performance, growth, resilience, internationalisation, and sustainability, but their effects depend on complementary resources, implementation capabilities, stakeholder incentives, and environmental conditions [27, 35].

Research on SME BMI has expanded substantially, but it remains distributed across several partly disconnected streams. Some studies explain BMI through resources and dynamic capabilities [29, 48], whereas others focus on entrepreneurial cognition, bricolage, effectuation, or strategic orientation [4, 7, 78]. A further stream examines networks and ecosystem relationships as sources of external knowledge, legitimacy, technologies, and complementary assets [41]. Digitalisation research investigates business model experimentation, platform participation, Industry 4.0 technologies, and digital capability reconfiguration [13, 51, 77], while sustainability research considers circular business models, stakeholder collaboration, and tensions between economic, environmental, and social objectives [8, 55, 62]. Recent studies further frame SME digital transformation as a dynamic capability process and the relationship between digital transformation and innovation as a multilevel phenomenon shaped by

organisational and environmental conditions [64, 65]. This diversity has generated valuable insights, but it has also produced conceptual fragmentation concerning what drives BMI, how it unfolds, what outcomes it produces, and under what conditions these relationships hold.

Several reviews address parts of this wider domain, but their analytical purposes differ from that of the present study. Miller et al. [47] establish the importance of treating SMEs as a heterogeneous organisational context rather than as scaled-down large firms, but their synthesis concerns business models broadly and is not a systematic review of SME BMI. Ancillai et al. [3] review digital technology and BMI across firms of different sizes and develop an explanatory framework for digitally enabled change, but they do not isolate the resource, managerial, and network mechanisms associated with SMEs. Zare and Persaud [81] map the wider intellectual structure of digital transformation and BMI through bibliometric analysis, yet their study does not treat SMEs as a distinct unit of analysis or examine the analytical roles and organisational mechanisms assigned to BMI in SME research. These reviews establish the importance of digitalisation in contemporary BMI research, but they leave scope for an SME-specific explanation of how digital technologies move from contextual inputs to substantive changes in value architecture.

Recent integrative reviews also demonstrate the value of applying an explicit organising lens when synthesising fragmented evidence. Latif et al. [40], for example, use a stage-contingent perspective to explain how the antecedents, processes, and outcomes of digital innovation vary across organisational stages. Although their substantive focus is digital innovation and female entrepreneurship rather than SME BMI, their approach illustrates how a review can move beyond descriptive categorisation by specifying classification rules, linking findings across analytical levels, reporting the strength and distribution of evidence, and deriving future research priorities from identifiable gaps. The present study adopts this general principle of explicit theoretical integration but does not apply Latif et al.'s stage-contingent framework. Instead, it uses the analytical role assigned to BMI and the mechanisms through which SME conditions are connected to business model change as its organising lens.

Accordingly, the central gap is not an absence of research on SME BMI. Rather, the existing evidence has not been integrated in a way that simultaneously clarifies: (i) the intellectual and methodological structure of the field; (ii) the different analytical positions assigned to BMI; (iii) the mechanisms connecting SME resource conditions, technologies, managerial action, and external relationships to BMI; (iv) the consequences and boundary

conditions of BMI; and (v) the strength of evidence supporting adaptive, iterative, or recursive interpretations of BMI. Existing bibliometric studies identify broad topics, while individual empirical studies commonly examine isolated antecedents or outcomes. What remains lacking is an SME-specific synthesis that connects thematic structure with article-level causal roles and clarifies how digitalisation, alongside other contextual conditions, is translated into business model reconfiguration through capability development and relational mechanisms.

To address this gap, the study asks the following research questions: What characteristics define the overall profile of existing literature on business model innovation in SMEs? How does the literature position BMI, and what mechanisms, tensions, and boundary conditions shape BMI in SMEs? What evidence gaps and research priorities should guide future research on SME BMI?

This study employs a systematic literature review methodology that integrates descriptive statistics, bibliometric co-word analysis [20, 53], and qualitative content analysis [33, 58] to comprehensively examine 162 SSCI-indexed journal articles. The keyword analysis maps the principal knowledge domains of the field, while the content analysis examines the dominant position assigned to BMI in each study's research question, analytical model, and findings. Articles are classified according to whether BMI is primarily treated as an outcome to be explained, an organisational process, an antecedent of subsequent outcomes, part of a combined causal relationship in which antecedent conditions influence BMI and BMI subsequently affects outcomes, or as a descriptive or typological construct. Longitudinal design, experimentation, iteration, and recursive feedback are coded separately to prevent cross-sectional mediation relationships from being interpreted as temporal feedback processes.

The keyword analysis reveals four broad knowledge domains: capabilities, BMI, and firm performance; SME business model design, entrepreneurship, and sustainability; digital transformation and technology-related antecedents; and strategy, knowledge management, and exploration. The qualitative synthesis shows that the literature assigns several distinct analytical roles to BMI rather than converging on a single causal position. Across these roles, a common explanatory pattern emerges: contextual pressures and opportunities influence BMI through entrepreneurial action logics, organisational capabilities, resource orchestration, knowledge absorption, experimentation, and network mobilisation. BMI may subsequently contribute to performance, resilience, growth, internationalisation, and sustainability, but these effects are conditional. In the digitalisation stream specifically, access to technologies or platforms is not sufficient; substantive DBMI depends on whether SMEs

can translate digital inputs into changes in their value architecture.

This study makes three contributions to the field of SME business model innovation. First, it provides an SME-specific integrative synthesis of a literature that has previously been dispersed across digitalisation, sustainability, entrepreneurship, capability, and network research. This domain focus clarifies how the organisational conditions of SMEs enter BMI explanations without assuming that SMEs possess universally distinctive or superior innovation processes. Second, the study provides analytical clarification by distinguishing the different roles assigned to BMI and separating combined causal models from evidence of temporal iteration and recursive feedback. This distinction prevents cross-sectional relationships from being overinterpreted as dynamic processes. Third, the review develops a contingent framework linking contextual conditions, organisational action, business model reconfiguration, outcomes, and boundary conditions. In doing so, it extends DBMI research by distinguishing digital technology adoption from substantive business model reconfiguration and by identifying the mechanisms and conditions through which digitally enabled BMI occurs.

Review methodology

Review design and scope

A systematic literature review was conducted to provide a transparent and reproducible synthesis of research on BMI in SMEs [56, 72]. During planning, preliminary reading indicated that existing reviews either examined SME business models broadly or focused on specific topics such as innovation capability, digitalisation, or particular sectors. The present review therefore focuses specifically on BMI as a focal construct in SMEs and integrates bibliometric mapping with qualitative content analysis.

Databases, search strategy, and eligibility criteria

Web of Science (WOS) and Scopus were selected because of their broad coverage of multidisciplinary bibliographic records [76, 84]. To apply a consistent journal-quality criterion across the two databases, all potentially eligible records, including those retrieved from Scopus, were manually checked against the December 2024 SSCI journal list [1, 44, 45]. The searches were conducted on 2 May 2025 using terms related to SMEs and BMI in the title, abstract, and keyword fields, as detailed in Table 1. The review was limited to English-language, peer-reviewed journal articles published in SSCI-indexed journals.

The following eligibility criteria were applied:

- (i) Studies had to focus on business model innovation in SMEs. Articles examining business models with-

Table 1 Search terms and Boolean combinations

Terms description	Keywords identified for the term
SMEs: According to European Commission [21], SMEs include three categories of enterprises: (i) micro-enterprises, (ii) small enterprises, and (iii) medium-sized enterprises. The search combined all the terms for these categories using the Boolean operator OR	SMEs: "SMEs" OR "SME" OR "small and medium enterprises" OR "small and medium-sized enterprises" OR "small and medium businesses" OR "small and medium-sized businesses" OR "small and medium companies" OR "small and medium sized companies" OR "micro small and medium enterprises" OR "micro small and medium sized enterprises" OR "micro-sized enterpris*" OR "micro compan*" OR "micro-sized compan*" OR "micro business*" OR "micro sized business*" OR "microfirm*" OR "micro-sized firm*" OR "Small enterprises: "small enterpris*" OR "small-sized enterpris*" OR "small compan*" OR "small-sized compan*" OR "small business*" OR "small sized business*" OR "small firm*" OR "small-sized firm*" OR "startup" OR "Medium-sized enterprises: "medium enterpris*" OR "medium-sized enterpris*" OR "medium compan*" OR "medium-sized compan*" OR "medium business*" OR "medium-sized business*" OR "medium firm*" OR "medium-sized firm*" OR "Business model inno*" OR "innov* business model" R "renew* business model" OR "business model renew*" OR "transform* business model" OR "business model transform"
BMI: The search used terms capturing business model innovation, renewal, and transformation, including variants previously used in the literature	

out a substantive BMI focus, or BMI exclusively in non-SME contexts, were excluded.

- (ii) BMI had to be a primary focus of the study, as determined through examination of the title, abstract, keywords, and full text. Studies mentioning BMI only tangentially were excluded.
- (iii) Only English-language journal articles published in SSCI-indexed journals were included. Edited books, conference proceedings, editorials, and articles published in non-SSCI journals were excluded.

Screening and sample construction

The searches identified 1074 records: 427 from Web of Science and 647 from Scopus. Applying the language and SSCI eligibility criteria excluded 734 records, leaving 340 potentially eligible records (217 from Web of Science and 123 from Scopus). We then removed 87 duplicate records, leaving 253 unique records for relevance screening. Assessment of the titles, abstracts, keywords, and full texts excluded a further 91 records that did not meet the inclusion criteria. The final review sample therefore comprised 162 articles (Fig. 1).

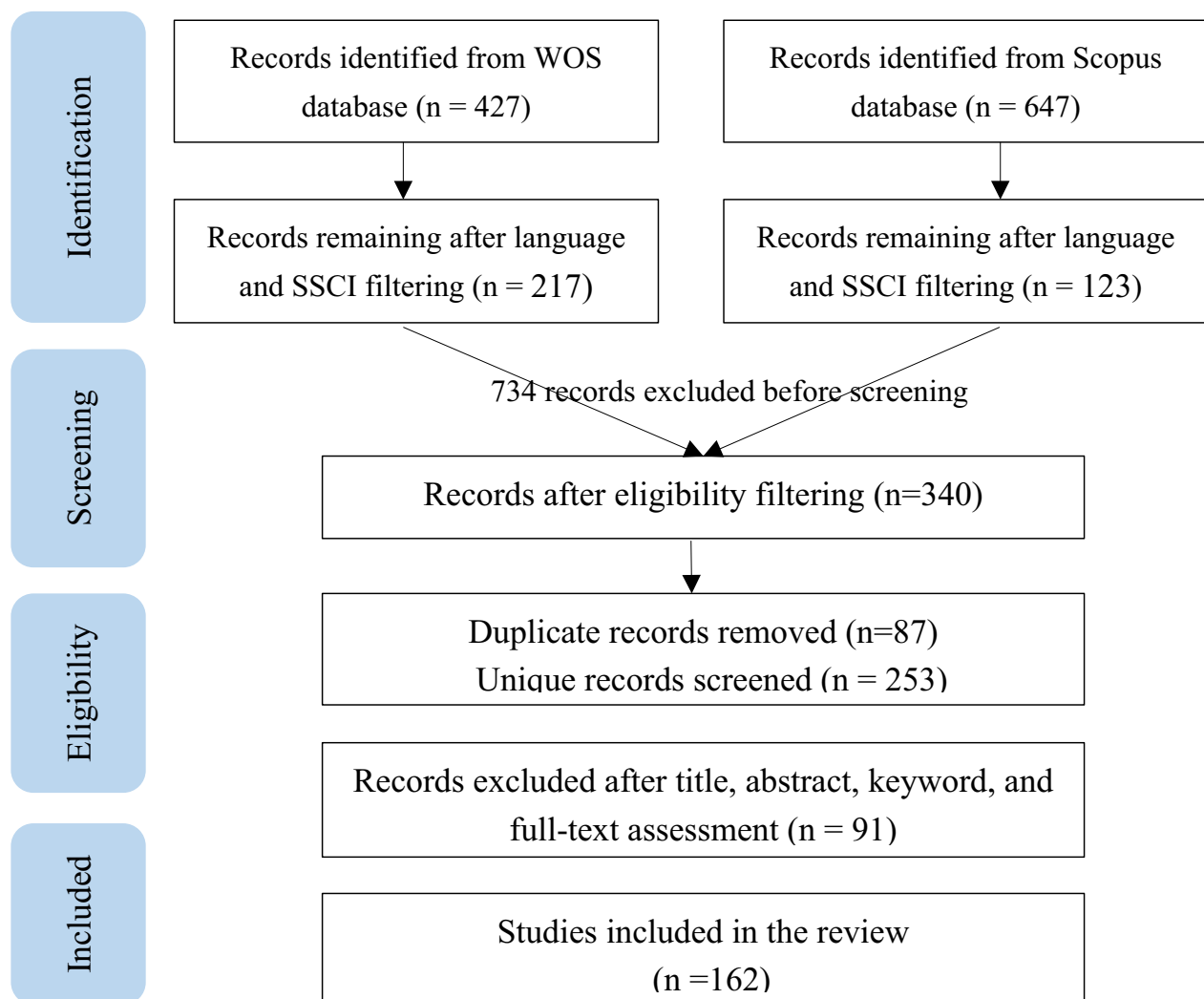
Data analysis

Descriptive analysis was used to examine publication years, journals, citation patterns, research designs, and theoretical perspectives. Keyword co-occurrence analysis was then conducted in VOSviewer version 1.6.20 using all keywords and the total counting method [14, 15, 67, 75]. The exported records contained 867 unique raw keywords. After synonymous terms, abbreviations, singular and plural forms, and spelling variants were consolidated, the minimum occurrence threshold was set at seven. Forty-five keywords met this threshold and were

included in the final network. VOSviewer grouped them into four clusters comprising 17, 15, 9, and 4 keywords.

Qualitative content analysis was conducted separately from the co-occurrence analysis. The unit of coding was the individual article. Each article received one mutually exclusive primary code according to the dominant position assigned to BMI in its research question, conceptual model, and principal findings. BMI was coded as an outcome when capabilities, technologies, networks, managerial factors, or contextual pressures were used to explain it; as a process when the study directly examined search, design, experimentation, implementation, adaptation, or temporal reconfiguration; as an antecedent when BMI was used to explain organisational or market outcomes; and as a combined causal role when an article examined a relationship in which antecedent conditions influenced BMI and BMI was subsequently linked to organisational or market outcomes.

Research design, temporal evidence, iteration, and recursive feedback were coded independently of the primary analytical role. A study was coded as recursive only when it provided explicit temporal or process evidence that performance outcomes, market responses, or stakeholder feedback informed a subsequent round of BMI. Cross-sectional mediation models were not treated as evidence of feedback. Coding decisions were documented in an article-level audit table and iteratively reviewed against abstracts and full texts. This procedure produced a traceable link between each synthesis claim, the articles supporting it, and the evidence limitations reported in Sect. 3.3. Studies published after the search cutoff or not meeting the corpus eligibility criteria were not retrospectively added to the coded sample. Where directly relevant, such studies are cited in the Discussion or theoretical contribution to position and interpret the review findings, but they do not

**Fig. 1** Summary of article selection

contribute to the bibliometric counts, article-level coding, or reported evidence frequencies.

Results and findings

Descriptive profile

The annual distribution shows that 120 of the 162 articles (74.1%) were published between 2021 and the search date of 2 May 2025 (Fig. 2). Publication output increased from 22 articles in 2021 and 22 in 2022 to 26 in 2023 and 37 in 2024; a further 13 articles had been published by the search date in 2025. Because 2025 is a partial publication year, it should not be compared directly with complete calendar years. The included articles had received 6621 citations in total when the dataset was exported.

Most publications are empirical (91%), with the remainder comprising conceptual or theoretical

work. Within the empirical subset, qualitative designs accounted for 58%, quantitative designs for 40%, and mixed methods for 2%. This distribution indicates a substantial qualitative knowledge base, but also highlights opportunities for longitudinal, comparative, and mixed-method research.

The articles were distributed across 67 journals. Table 2 reports the 21 journals that published at least three articles; the remaining 54 articles appeared across 46 journals that published one or two articles each. Technological Forecasting and Social Change had the highest average citation count among the listed journals (106.4 citations per article), followed by Journal of Business Research (105.3). Sustainability published the largest number of articles (15) but had a lower average citation count (17.7). Among recurring authors, Müller had the highest citation total, with three articles receiving 865

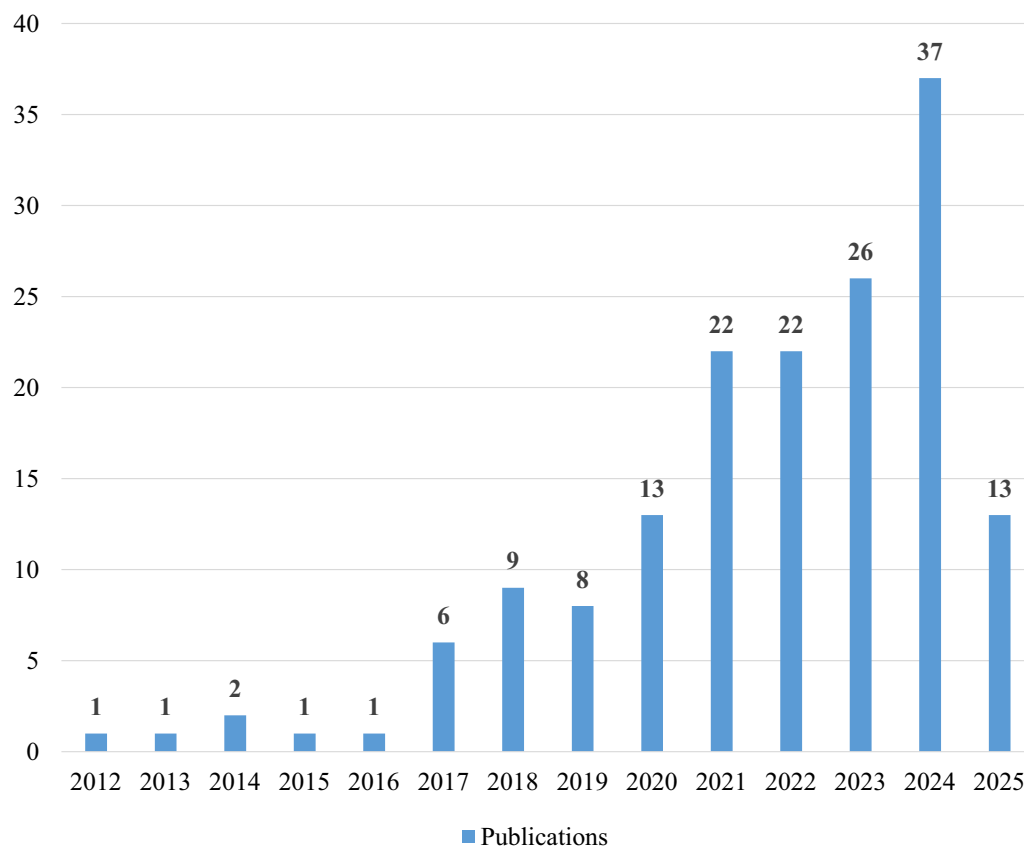


Fig. 2 Annual distribution of the publications included in the review

citations; Kraus and Bouwman each contributed five articles, receiving 693 and 477 citations, respectively.

The analysis of the 50 most-cited articles identified dynamic capabilities and the resource-based view as the most frequently used theoretical perspectives (Table 3). Service-dominant logic, value co-creation, open innovation, institutional theory, contingency theory, and resource orchestration were also represented. Multiple theories were often combined within a single article, indicating that SME BMI is commonly explained through complementary resource, capability, relational, and environmental perspectives rather than one dominant theory.

Keyword co-occurrence analysis

Forty-five keywords met the minimum occurrence threshold of seven and were included in the final co-occurrence network. VOSviewer assigned these keywords to four clusters: red ($n = 17$), green ($n = 15$), blue ($n = 9$), and yellow ($n = 4$). Figure 3 presents the network map, while Table 4 reports the occurrence frequency and total link strength of each keyword. The cluster labels were assigned by the authors after examining the most frequent and strongly connected keywords and

cross-checking their interpretation against representative studies in the reviewed corpus. The labels therefore summarise patterns of topical association rather than causal relationships.

The red cluster is labelled “Capabilities, BMI, and firm performance”. It is anchored by business model innovation (113 occurrences), firm performance (88), dynamic capabilities (26), entrepreneurial orientation (24), value creation (19), market orientation (13), and absorptive capacity (8). Together, these terms indicate a research stream that examines how organisational and entrepreneurial capabilities are associated with the performance implications of BMI, frequently through mediating and moderating relationships. Representative studies consider the roles of digital capabilities, dynamic capabilities, knowledge management, and strategic flexibility in linking BMI with firm and international performance [5, 28, 31, 48].

The green cluster is labelled “SME business model design, entrepreneurship, and sustainability”. Its most prominent terms include SMEs (56), innovation (33), business model (26), design (24), management (24), entrepreneurship (23), sustainability (20), and circular

Table 2 Journals publishing three or more articles in the review sample

Journal	Articles	Total citations	Average citations per article
Technological Forecasting and Social Change	7	745	106.4
Journal of Business Research	8	842	105.3
Journal of Cleaner Production	5	454	90.8
International Entrepreneurship and Management Journal	3	270	90.0
R&D Management	4	351	87.8
Creativity and Innovation Management	5	337	67.4
Long Range Planning	3	199	66.3
Technovation	5	283	56.6
Journal of Innovation & Knowledge	3	158	52.7
Journal of Manufacturing Technology Management	5	251	50.2
Business Strategy and the Environment	5	194	38.8
Management Decision	3	108	36.0
Technology Analysis & Strategic Management	6	146	24.3
Journal of the Knowledge Economy	5	101	20.2
Sustainability	15	266	17.7
Journal of Knowledge Management	3	53	17.7
IEEE Transactions on Engineering Management	7	114	16.3
Journal of Small Business Management	4	56	14.0
Frontiers in Psychology	3	38	12.7
European Journal of Innovation Management	6	61	10.2
Business Process Management Journal	3	7	2.3
Others	54	1587	29.4
Total	162	6621	40.9

Table 3 Theoretical perspectives used in the 50 most-cited articles

Theory/perspective	Number of articles using the theory
Dynamic capabilities	11
Resource-based view	10
Service-dominant logic (perspective)	5
Value co-creation theory (Perspective)	5
Open innovation theory	4
Institutional theory	4
Contingency theory	3
Resource orchestration theory	2
Others	1 each

Multiple coding was permitted; therefore, the frequencies sum to 60 theoretical applications across the 50 articles. The “Others” category includes the following theories and perspectives, each coded once: Emergent Strategy Theory, Entrepreneurial Alertness Theory, Entrepreneurial Ecosystem Theory, Entrepreneurial Orientation Theory, Equifinality Theory, Exaptation Theory, Intellectual Capital Theory, Leadership and Organizational Knowledge Creation Perspective, Network Ties Perspective, Social Capital Theory, Social Exchange Theory, Stakeholder Theory, Strategic Leadership Theory, Trait Activation Theory, and Upper Echelons Theory

economy (12). This configuration reflects a broad stream concerned with the design and management of SME business models, entrepreneurial opportunities and challenges, and the incorporation of sustainability and circularity. Representative studies examine sustainable business model development, circular transformation, and the ecosystem and institutional conditions that support or constrain implementation in SMEs [55, 57, 62].

The blue cluster is labelled “Digital transformation and technology-related antecedents”. It contains technologies (29), antecedents (14), digital transformation (11), industry (11), orientation (11), transformation (10), creation (9), product (8), and uncertainty (7). These terms point to research on the technological and environmental conditions associated with BMI, including digital transformation pressures, industry context, strategic orientation, product change, and uncertainty. Representative studies examine how digitalisation and Industry 4.0 technologies enable or require changes in SME business models and value creation processes [13, 46, 50].

The yellow cluster, the smallest of the four, is labelled “Strategy, knowledge management, and exploration”. It

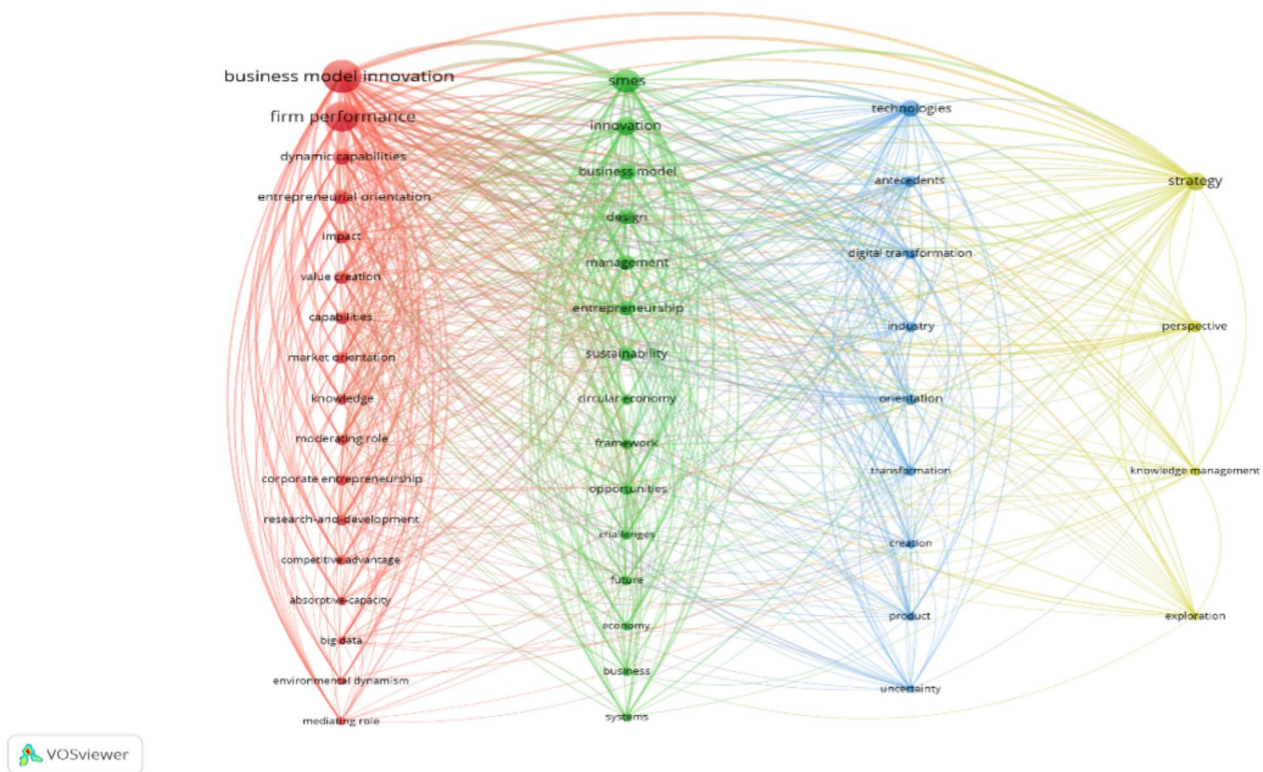


Fig. 3 Keyword co-occurrence analysis: network map

comprises strategy (32), perspective (13), knowledge management (8), and exploration (7). Although compact, the cluster identifies an interdisciplinary stream concerned with strategic interpretation, external and internal knowledge management, and exploratory approaches to BMI. Representative studies consider how knowledge management, absorptive capacity, entrepreneurial networks, and effectual or bricolage-based action support BMI under resource constraints and uncertainty [9, 31, 79, 80].

Qualitative content analysis: analytical roles assigned to BMI

The content analysis examines the position assigned to BMI in each article's research question, analytical model, and principal findings. Of the 162 articles, 138 assigned BMI a focal analytical role: 17 treated BMI primarily as an antecedent, 41 as an outcome, 44 as an organisational process, and 36 as part of a combined causal relationship. The remaining 24 articles were descriptive, conceptual, or typological and did not assign BMI a dominant causal position. The coexistence of these roles across the literature does not demonstrate that an individual SME performs all of them simultaneously.

Temporal and process evidence was coded separately. Eleven studies used explicitly longitudinal designs, and 18 documented iteration, experimentation, or repeated reconfiguration without explicit feedback. These categories were coded independently and may overlap; they should not be interpreted as 29 distinct studies. Direct feedback was coded more conservatively and required explicit evidence that earlier outcomes, market responses, or stakeholder reactions informed later BMI activity. This distinction prevents repeated change or cross-sectional mediation from being treated as recursive feedback.

Distribution and analytical scope

Process studies form the largest category, followed by studies positioning BMI as an outcome and studies examining a combined causal relationship. The distribution indicates that the field is concerned with what drives BMI, how BMI develops, what it produces, and how it connects organisational conditions with subsequent outcomes. The 24 descriptive and typological studies contribute business model archetypes, design configurations, measurement approaches, and emerging digital or circular forms, but they are not used to infer causal direction or temporal sequence [30, 63, 73]. Table 5 summarises

Table 4 Keyword co-occurrence clusters

	Keyword occurrences	Total link strength
Red cluster ($n = 17$): Capabilities, BMI, and firm performance		
Business model innovation	113	521
Firm performance	88	476
Dynamic capabilities	26	156
Entrepreneurial orientation	24	133
Impact	19	117
Value creation	19	105
Capabilities	16	88
Market orientation	13	83
Knowledge	12	74
Moderating role	12	69
Corporate entrepreneurship	11	61
Research-and-development	11	71
Competitive advantage	10	57
Absorptive-capacity	8	50
Big data	7	40
Environmental dynamism	7	33
Mediating role	7	39
Green cluster ($n = 15$): SME business model design, entrepreneurship, and sustainability		
SMEs	56	326
Innovation	33	178
Business model	26	138
Design	24	147
Management	24	130
Entrepreneurship	23	140
Sustainability	20	97
Circular economy	12	43
Framework	12	65
Opportunities	12	64
Challenges	10	61
Future	10	61
Economy	9	38
Business	8	37
Systems	8	50
Blue cluster ($n = 9$): Digital transformation and technology-related antecedents		
Technologies	29	159
Antecedents	14	79
Digital transformation	11	68
Industry	11	57
Orientation	11	59
Transformation	10	51
Creation	9	46
Product	8	43
Uncertainty	7	42
Yellow cluster ($n = 4$): Strategy, knowledge management, and exploration		
Strategy	32	182
Perspective	13	79
Knowledge management	8	45
Exploration	7	42

Table 5 Analytical roles assigned to BMI in the reviewed literature

Category	Decision rule	N (%)	Dominant focus	Representative studies	Main evidence limitation
BMI as outcome	Capabilities, technologies, networks, managerial factors, or contextual presures explain BMI	41 (25.3%)	Knowledge and absorptive capacity; dynamic and digital capabilities; leadership; networks; institutions	Hock-Doeppgen et al. [31], Miroshnychenko et al. [48], Xie et al. [77], Xu, He, et al. (79, 80)	Many studies are cross-sectional and treat BMI as an aggregate construct
BMI as process	The study directly examines search, design, experimentation, implementation, adaptation, or temporal reconfiguration	44 (27.2%)	Search and experimentation; capability deployment; parallel models; temporary BMI; circular transformation	Andersen et al. [4], Bitetti and Gibbert [10], Clauss et al. [18], Paiola et al. [54]	Only 11 studies used explicitly longitudinal designs; retrospective accounts limit claims about process prevalence; retrospective accounts limit prevalence claims
BMI as antecedent	BMI predicts organisational, market, customer, sustainability, or performance outcomes	17 (10.5%)	Performance; growth; internationalisation; sustainability; customer process innovation; survival	Bouwman et al. [13], Cucculelli and Bettinelli [19], Jean et al. [36], Rehman et al. [61]	Effects depend on implementation, complementary assets, incentives, and context
Combined causal role	Antecedent conditions influence BMI, which subsequently affects one or more outcomes	36 (22.2%)	Mediation and linked antecedent–BMI–outcome models	Guo et al. [27], Ferreras-Méndez et al. [23], Lew et al. [41], Anwar et al. [5]	Cross-sectional mediation does not demonstrate temporal recursion or feedback
Descriptive or typological	The study describes, classifies, conceptualises, or measures business models without assigning BMI a dominant causal position	24 (14.8%)	Archetypes; design configurations; conceptual models; measurement; bibliometric mapping	Henry et al. [30], Richter et al. [63], Trimi and Berbegal-Mirabent [73]	Useful for construct development, but limited for causal and process inference

the coding rules, distribution, dominant focus, representative evidence, and principal limitations of the five groups.

BMI as an outcome: drivers and enabling conditions

Forty-one studies treated BMI primarily as an outcome to be explained. Four groups of antecedents recur: organisational capabilities and knowledge, digital technologies and infrastructure, managerial cognition and strategic orientation, and external networks or institutional conditions. Across these groups, access to a resource or technology is rarely sufficient by itself. BMI depends on whether SMEs can interpret, combine, deploy, and govern the resources available to them.

Capability-based studies distinguish among mechanisms rather than treating capability as an aggregate resource. External knowledge management capabilities stimulate BMI more consistently than internal knowledge alone, while the contribution of internal knowledge varies with risk tolerance [31]. Potential and realised absorptive capacity performs different functions, and the effect of realised absorptive capacity on BMI becomes stronger under environmental uncertainty [48]. Dynamic capability research further indicates that different capabilities support different business model components [29]. The shared mechanism is resource transformation: knowledge and capabilities influence BMI when they enable firms to recognise, assimilate, and redeploy resources into a revised value logic.

Digitalisation studies reach a similar conclusion. Platforms, infrastructure, and technologies expand possible business model configurations, but their influence is mediated by capability reconfiguration, complementary assets, organisational culture, and routine updating [42, 77, 82]. Digital investment is therefore an input into organisational change, not evidence that BMI has already occurred. Managerial and relational studies add further contingencies. Executive cognition may influence BMI through entrepreneurial bricolage, and this relationship varies with environmental dynamism [32]. CEO characteristics, failure experience, managerial persistence, and financial resource slack also shape the likelihood or extent of BMI [7, 52]. Networks supply knowledge, legitimacy, partners, and market access, but their effects depend on effectual action, absorptive capacity, and the firm's ability to govern external relationships [78–80]. BMI is therefore a contingent organisational achievement rather than an automatic response to technology or scarcity.

BMI as a process: how BMI unfolds

Forty-four studies examined BMI as a process of search, design, experimentation, implementation, adaptation, or reconfiguration. They agree that BMI is rarely a single decision. It involves interconnected activities through which SMEs interpret opportunities and threats, evaluate alternatives, mobilise resources, and modify value creation and capture arrangements. They do not identify one universal sequence.

One stream emphasises organised search and experimentation. SMEs use search behaviours, prototypes, staged commitment, and alternative configurations because the viability of a new model cannot be fully specified in advance [4, 49]. In digital settings, old and new models may operate in parallel while firms learn how connected products, services, and ecosystem relationships alter their value logic [54]. A second stream explains BMI through capabilities and managerial interpretation. Longitudinal studies show how market orientation, strategic agility, sensing, and dynamic capabilities support repeated renewal by shaping how managers notice change, frame alternatives, commit resources, and coordinate reconfiguration [6, 10, 60].

Crisis and sustainability research reveals a more emergent and contested process. Temporary BMI may help SMEs redeploy existing knowledge, competences, and relationships during disruption, but the resulting configurations may later be abandoned, adapted, or incorporated into the established model [11, 18, 26]. Circular BMI also requires firms to manage tensions between economic viability, environmental objectives, organisational routines, and stakeholder expectations [55]. Evidence from the longitudinal subset and from studies documenting iteration, experimentation, or repeated reconfiguration supports an adaptive and frequently iterative interpretation of SME BMI. Direct feedback evidence is much more limited; the clearest example uses dynamic business modelling to identify reinforcing and balancing feedback loops [11]. These findings do not establish that all SMEs follow the same pathway or that recursion is universal.

BMI as an antecedent: consequences and contingencies

Seventeen studies primarily positioned BMI as an antecedent of organisational or market outcomes, including performance, growth, internationalisation, organisational sustainability, sustainable value creation, customer process innovation, and survival. The common premise is that revising the firm's value logic can improve its ability to exploit opportunities or respond to changing conditions. The evidence nevertheless qualifies a simple positive relationship.

The reviewed evidence suggests that BMI is more likely to contribute to favourable outcomes when it is supported by complementary assets and effective implementation. Business model change was more strongly associated with performance when combined with intangible investment [19]. Experimentation and strategy implementation improved the performance of digitalising SMEs through organisational innovativeness rather than through digitalisation alone [13]. International performance effects vary with entrepreneurial orientation and institutional conditions [36]. Circular and sustainable models can improve economic, environmental, and social outcomes, but the result depends on incentives, stakeholder alignment, customer behaviour, and institutional support [12, 61]. Similarly, Industry 4.0 solution providers may use BMI to enable process innovation for customers, but the benefits depend on the solution offered and the customer's capacity to adopt it [51]. BMI should therefore be understood as a potential mechanism for value creation rather than an automatically beneficial intervention.

BMI in a combined causal role

Thirty-six studies placed BMI within models in which antecedent conditions were theorised to influence BMI and BMI was subsequently linked to organisational or market outcomes. Typical models connect opportunity recognition, entrepreneurial orientation, digital capabilities, network ties, intellectual capital, managerial alertness, or sustainability orientation to BMI and then to performance, growth, internationalisation, competitiveness, resilience, or sustainability [5, 23, 27, 41].

This stream identifies BMI as an explanatory bridge. Opportunity recognition contributes to performance when firms translate an identified opportunity into a revised model [27]. Entrepreneurial orientation affects new product development through BMI [23], while digital capabilities contribute to internationalisation and competitiveness partly through business model innovativeness [5, 69]. Network ties and internal capabilities similarly influence growth or venture success through BMI [74, 83]. However, a combined causal model is not equivalent to a recursive process. Cross-sectional survey data and mediation analysis can support a theoretically ordered relationship in which antecedent conditions influence BMI and BMI is subsequently linked to outcomes, but they do not show whether outcomes or stakeholder responses trigger later rounds of BMI. Claims of feedback are therefore reserved for studies that provide explicit temporal or process evidence.

Cross-stream synthesis and unresolved tensions

The synthesis yields four conclusions. First, the literature does not converge on one causal position for BMI. Some studies explain what produces BMI, some examine how it unfolds, some investigate what it produces, and others connect these relationships within a linked causal model. This plurality reflects different research questions and designs, not simultaneous behaviour within every SME.

Second, the streams converge on a mechanism of resource mobilisation and value reconfiguration. Contextual pressures and opportunities are interpreted through entrepreneurial action logics and translated through absorptive capacity, dynamic capabilities, resource orchestration, experimentation, and network mobilisation into changes in value proposition, value creation and delivery, and value capture. External networks are especially important, but externally sourced knowledge and resources must be absorbed, recombined, and governed.

Third, the evidence challenges an unqualified resource-constraint advantage. Moderate constraints may encourage bricolage, prioritisation, and focused experimentation, whereas severe financial, skill, and managerial shortages can delay or prevent BMI. Digitalisation may improve connectivity, efficiency, and market access while creating investment costs, capability requirements, implementation risks, and platform dependence. Sustainability-oriented BMI may generate long-term environmental and social value while increasing short-term financial and organisational demands.

Fourth, temporal evidence supports adaptation and iteration more strongly than recursion. A small subset of longitudinal and process studies suggests that market responses, performance outcomes, or stakeholder feedback may inform subsequent experimentation and reconfiguration. SME BMI can therefore be characterised as frequently adaptive and iterative, while recursive feedback is documented only in a limited subset of studies and contexts.

Further discussion

The combined analyses provide two complementary views of the field. Keyword co-occurrence analysis maps the topics around which SME BMI research has accumulated, while the role-based content analysis explains how BMI is positioned and what evidence supports the proposed relationships. The discussion therefore moves beyond the four keyword clusters and integrates the findings around resource mobilisation, organisational and relational mechanisms, digital-sustainability tensions, and temporal adaptation.

From resource constraints to resource mobilisation

A central implication is that resource constraints should not be treated as either uniformly detrimental or inherently beneficial. The reviewed evidence instead indicates conditional effects. Costs and resource scarcity may trigger business model redesign, bricolage, and focused experimentation, but their effects depend on environmental uncertainty, financial slack, and access to complementary resources [16, 52, 79, 80]. CircularBMI studies also identify financial, legal, organisational, and value-chain barriers that can restrict implementation [25, 71]. The theoretically important issue is therefore not scarcity itself, but the mechanisms through which SMEs mobilise, supplement, and reconfigure constrained resources.

This interpretation shifts attention from the existence of resource constraints to the actions through which constrained resources are interpreted, supplemented, and recombined. Bricolage, effectuation, external knowledge mobilisation, and capability reconfiguration provide different explanations of this translation process [32, 78–80]. Their specific functions and relationships are examined in the following section.

Mechanisms linking SME conditions to BMI

The review identifies a recurring explanatory pattern. Contextual conditions create pressures and opportunities, entrepreneurial action logics shape how managers interpret them, and organisational and relational mechanisms determine whether they are converted into BMI. Absorptive capacity enables external knowledge to be recognised and transformed; dynamic capabilities support sensing, commitment, and reconfiguration; resource orchestration coordinates available assets; and networks provide access to complementary knowledge, legitimacy, technology, and markets [31, 48, 78]. Complementary research outside the coded corpus reinforces this interpretation by framing SME digital transformation as a dynamic capability process and the digital transformation–innovation nexus as a multilevel phenomenon, while also suggesting that ecosystem coopetition may contribute to sustainable innovation through digital transformation and resource orchestration [37, 64, 65].

This pattern explains why technology, networks, and managerial orientation often have indirect or contingent effects. Digital platforms influence BMI through capability reconfiguration and complementary assets, rather than substituting for business model change [42, 77]. Digital transformation and digital capabilities contribute to performance or internationalisation when they are translated through BMI [5, 13, 46]. Entrepreneurial networks likewise influence BMI through effectuation, causation, or bricolage, and these relationships vary with environmental dynamism [78, 78, 80]. The review therefore shifts

explanation away from lists of antecedents and towards the mechanisms that connect conditions to changes in the business model architecture.

Digital and sustainability synergies and tensions

Digitalisation and sustainability are frequently presented as mutually reinforcing, but the reviewed evidence shows that complementarity is mediated rather than automatic. Digital platforms and digitalisation can support BMI through capability reconfiguration, complementary assets, organisational routine updating, and servitisation strategies [17, 42, 77, 82]. Complementary research outside the coded corpus indicates that digital transformation may support sustainability innovation and performance through enhanced information-processing capacity and sustainability innovation ambidexterity [43]. Configurational evidence further suggests that sustainability outcomes depend on combinations of digital transformation and environmental innovation rather than on either condition operating independently [59].

Sustainability-oriented BMI likewise combines opportunity and constraint. Pay-per-use models can change consumption patterns when environmental outcomes are intentionally designed into the business model [12], while circular economy innovation and BMI can contribute to economic, environmental, and social performance when government incentives are available [61]. At the same time, circular BMI faces financial, legal, organisational, and value-chain barriers, as well as paradoxical tensions between competing priorities and goals [25, 55, 71]. Ecosystem strategies are also contingent on the role adopted by the SME and the maturity of the surrounding ecosystem [62]. Digitalisation and sustainability should therefore be analysed as conditionally complementary: their joint value depends on capabilities, complementary assets, financing, incentives, ecosystem configuration, and institutional support.

Adaptive and iterative BMI, with limited evidence of recursion

Adaptation refers to business model adjustment in response to changing conditions. Iteration involves repeated experimentation, trial-and-error learning, or redesign. Recursion requires stronger evidence that the outcomes of an earlier business model change or stakeholder response trigger a later round of BMI. Representative studies trace strategic agility and dynamic capabilities over time, organised search and experimentation, parallel business models, and trial-and-error learning [4, 6, 10, 54, 60]. Taken together, the evidence is considerably stronger for adaptation and iteration than for recursion. This uneven evidence base suggests that repeated business model adjustment should not automatically be interpreted as a

feedback process in which prior outcomes initiate subsequent rounds of BMI.

This distinction resolves the tension between static causal models and process accounts. Cross-sectional mediation models are useful for explaining how antecedents may operate through BMI, but they cannot establish temporal feedback. Future theory should therefore specify when one-directional BMI processes develop into recursive processes and which organisational or environmental conditions sustain subsequent rounds of change.

An integrative framework of contingent SME BMI

Figure 4 synthesises the recurrent relationships identified in the reviewed corpus into a contingent framework. Contextual conditions include resource constraints, digital pressures and opportunities, sustainability demands, and environmental or institutional uncertainty. These conditions influence BMI through entrepreneurial interpretation and organisational and relational mechanisms that support changes in value proposition, value creation and delivery, and value capture. In relation to digitally enabled BMI, digital pressures and opportunities enter the framework as contextual conditions rather than as business model innovation in themselves. They become relevant to BMI when firms translate digital inputs into substantive changes in their value architecture [3, 77]. Business model reconfiguration may subsequently contribute to performance, growth, internationalisation, resilience, survival, and sustainability, although these outcomes are not automatic.

The framework does not assume that every relationship is positive or that all SMEs follow the same sequence.

Financial slack, complementary assets, environmental uncertainty, organisational routines, ecosystem roles, and institutional support alter whether contextual conditions can be translated into BMI and whether BMI produces desired outcomes [42, 48, 52, 57, 62, 82]. The dashed feedback relationship therefore represents an emerging possibility rather than a generally established relationship. Figure 4 is the authors' synthesis of recurring relationships and evidence boundaries. It does not convert bibliometric clusters into causal determinants or treat recursion as a defining characteristic of all SMEs.

Conclusion and implications

Summary of the findings

This review synthesises 162 articles to clarify how business model innovation is conceptualised, explained, and examined in the SME literature. Its central conclusion is that SME BMI cannot be understood through a single causal role or a simple relationship between resource conditions and organisational outcomes. Across the literature, BMI appears as an outcome of managerial, technological, organisational, and relational conditions; as a process of search, experimentation, implementation, and reconfiguration; and as a mechanism through which firms pursue performance, growth, resilience, internationalisation, and sustainability. These perspectives are complementary at the level of the literature, but they should not be interpreted as a universal sequence enacted by every SME. Our evidence supports an adaptive and iterative interpretation of SME BMI, while direct evidence of recursive feedback remains limited. The proposed framework should therefore be understood as a

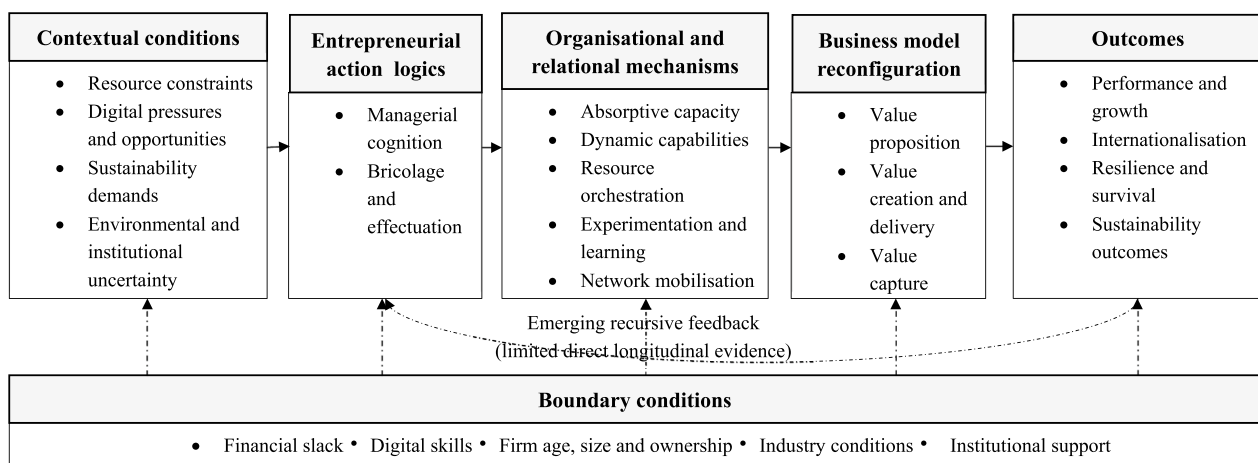


Fig. 4 Integrative framework of contingent SME BMI. *Source:* Authors' synthesis based on the reviewed literature. The framework represents an analytical sequence rather than a universal linear process. Solid arrows indicate recurrent relationships across the reviewed literature. Dotted connectors indicate boundary conditions that enable or constrain these relationships. The dashed feedback arrow denotes a limited pattern documented in a small subset of longitudinal studies

contingent synthesis rather than a universally validated causal sequence. It provides a basis for examining when SMEs can translate digitalisation, resource constraints, networks, and sustainability demands into viable changes in how they create, deliver, and capture value.

Theoretical implications

This review expands the current conversation on BMI in three related ways. First, it provides a domain-specific integration of SME BMI research. Existing reviews have examined SME business models broadly or synthesised digital technology, digital transformation, and BMI across heterogeneous firm populations [3, 47, 81]. The present review brings together digitalisation, sustainability, capability, entrepreneurship, and network research within the specific organisational conditions of SMEs. It therefore clarifies how resource limitations, concentrated managerial decision-making, organisational flexibility, and dependence on external relationships enter explanations of BMI without assuming that SMEs possess universally distinctive or superior innovation processes.

Second, the review provides analytical clarification by distinguishing BMI as an outcome to be explained, an organisational process, an antecedent of subsequent outcomes, and part of a combined causal relationship in which antecedent conditions influence BMI and BMI subsequently affects outcomes. This classification separates questions concerning what produces BMI, how BMI unfolds, and what BMI subsequently produces. It also distinguishes cross-sectional causal ordering from temporal development, experimentation, and recursive feedback. The review thereby prevents mediation relationships or the coexistence of different scholarly perspectives from being interpreted as evidence that individual SMEs necessarily enact all BMI roles simultaneously or follow a universally recursive process.

Third, the integrative framework extends DBMI research by shifting attention from whether digital technologies influence BMI to how and under what conditions digital inputs are translated into changes in a firm's value architecture. By locating business model reconfiguration between digital pressures and subsequent outcomes, the framework explains why technology adoption may improve operations without altering the underlying business model and why its performance and sustainability implications depend on capabilities, complementary assets, organisational routines, ecosystem position, and institutional support. The contribution therefore moves DBMI research beyond identifying digital antecedents towards explaining the mechanisms and boundary conditions through which digitalisation contributes to substantive business model innovation.

Managerial and policy implications

SME managers should distinguish technology adoption from digitally enabled BMI. A digital platform, analytics system, connected product, or automated process may improve operational efficiency without changing the underlying business model. Digital investments become strategically relevant to BMI when they are combined with experimentation, capability reconfiguration, complementary assets, and organisational routine updating and when they support substantive changes in value proposition, value creation and delivery, or value capture [3, 13, 42, 77]. Managers should therefore evaluate digital investments not only through adoption and efficiency indicators, but also through their contribution to changes in the firm's value architecture.

Networks can provide knowledge, partners, legitimacy, and complementary resources, but SMEs must actively absorb, coordinate, and govern these resources. Effectuation, causation, bricolage, absorptive capacity, and ecosystem roles influence whether external relationships are translated into BMI [62, 8078 –]. Managers should therefore assess not only the availability of external resources, but also their firm's capacity to integrate them and manage potential dependence, coordination costs, and bargaining asymmetries. Sustainability-oriented BMI likewise requires deliberate design and implementation. Circular, pay-per-use, and sustainability-oriented models may generate environmental, social, and economic value, but their outcomes depend on access to finance, customer acceptance, stakeholder alignment, organisational capability, and the management of competing priorities [12, 55, 71]. Sustainability objectives should therefore be integrated into the business model architecture rather than treated as an assumed consequence of digitalisation or innovation.

Policy support should move beyond generic technology adoption programmes. Evidence concerning public procurement and government incentives suggests that institutional support can reduce some financial and implementation barriers and strengthen the economic, environmental, and social outcomes associated with innovative business models [57, 61]. Effective interventions may therefore combine access to finance with capability development, experimentation support, and ecosystem brokerage. Because SMEs differ in size, age, ownership, sector, and region, policy design should be contingent rather than based on a uniform model of SME innovation. Support mechanisms should be adapted to the specific constraints preventing firms from translating technologies, external relationships, and sustainability pressures into viable business model reconfiguration.

Table 6 Evidence-based future research agenda

Evidence gap identified	Future research question	Suggested method	Expected contribution	Indicative evidence
Temporal evolution and feedback	When and how do experiments become sustained BMI, and when do outcomes trigger subsequent rounds of change?	Longitudinal multiple-case studies; panel designs; process tracing; event-sequence analysis	Establish temporal mechanisms and distinguish repeated adjustment from recursive feedback	Arbussa et al. [6], Randhawa et al. [60], Andersen et al. [4], Paiola et al. [54], Bitetti and Gibbert [10], Bivona and Cruz [11]
Digital adoption and substantive DBMI	Under what conditions do digital technologies remain operational improvements, and when do they lead to changes in value proposition, value creation and delivery, or value capture?	Longitudinal case studies; process tracing; multilevel designs; studies of failed or partial digital transformation	Distinguish technology adoption from DBMI and identify the mechanisms and boundary conditions through which digital inputs are translated into business model reconfiguration	Bouwman et al. [13], Xie et al. [77], Li et al. [42], Merin-Rodríguez et al. [46], Zhou et al. [82]
Threshold effects of resource constraints	When do costs and resource constraints stimulate redesign and bricolage, and when do they prevent substantive BMI?	Configurational analysis; matched cases; longitudinal resource measures	Replace a general constraint-advantage assumption with threshold and contingency explanations	Chammassian and Sabatier [16], Nyuur et al. [52], Xu He et al. [79], Geissdoerfer et al. [25], Toxopeus et al. [71]
Tensions between digitalisation and sustainability	Under what conditions do digitalisation and sustainability reinforce or constrain each other in SME BMI?	Multilevel studies; fsQCA; comparative case research; field experiments	Explain capability complementarities, financing barriers, incentives, and circular BMI tensions	Bocken et al. [12], Rehman et al. [61], Geissdoerfer et al. [25], Pascucci et al. [55]
Network governance and dependence	When do ecosystems provide complementary resources, and when do governance problems or dependence limit BMI?	Dyadic and ecosystem studies; network analysis; governance comparisons	Extend network research beyond its present emphasis on resource access and positive effects	Xu et al. [78], Xu, Wu et al. [80], Lew et al. [41], Reim et al. [62]
Comparison between SMEs and large firms	Are adaptive and iterative BMI mechanisms more prevalent or differently configured in SMEs than in larger firms?	Matched comparative samples; cross-size case designs; multi-group longitudinal models	Provide direct evidence for size-contingent claims	Miller et al. [47]
Institutional and contextual differences	How do ownership, age, sector, geography, environmental uncertainty, and institutional support alter BMI mechanisms and outcomes?	Cross-country panels; multilevel models; comparative institutional analysis	Specify boundary conditions and improve transferability	Miroshnichenko et al. [48], Peñate-Valentín et al. [57], Rehman et al. [61], Jean et al. [36]

Limitations and future research agenda

This study is also subject to several limitations. First, the search was restricted to English-language articles in SSCI-indexed journals retrieved from Web of Science and Scopus. Relevant work in other languages, non-SSCI journals, books, conference proceedings, and practitioner sources may therefore be underrepresented. Second, search terms and the requirement that BMI be a focal construct may have excluded studies that examine substantively similar change using different terminology. Third, keyword co-occurrence analysis maps associations among terms but cannot establish causal relationships. Cluster labels are interpretive summaries based on keyword frequency, link strength, and representative studies, and alternative labels may also be defensible. Fourth, the article-level role classification requires judgement about each study's dominant analytical focus. Although decisions were documented and ambiguous cases were re-examined, another coding scheme could produce somewhat different category boundaries. Fifth, the underlying evidence base limits the claims that can be made. Many studies rely on cross-sectional or retrospective designs, only 11 are explicitly longitudinal, and direct feedback evidence is substantially more limited than evidence of experimentation or repeated reconfiguration. The review therefore cannot establish the prevalence or causal sequence of adaptive and recursive BMI across the SME population. Sixth, because the corpus does not contain a systematically matched sample of large-firm studies, the review cannot directly demonstrate that the identified patterns are unique to SMEs. Finally, the proposed framework is an integrative synthesis and has not yet been independently tested as a complete model.

The future research agenda is derived from the evidence gaps identified in the review rather than from the keyword clusters alone. Table 6 connects each gap with a focused research question and an appropriate methodological response. Longitudinal research is the highest priority. Multi-wave case studies, process tracing, panel data, and event-sequence analysis are needed to identify when experimentation becomes sustained BMI and when outcomes initiate subsequent rounds of change. Direct comparative designs should examine whether temporal patterns, resource mobilisation mechanisms, and network dependence differ systematically between SMEs and larger firms. Research should also examine digital and sustainability tensions rather than assuming synergy. Studies can investigate when digital investment improves circular or sustainable value creation, when it produces cost and capability burdens, and how platform governance affects SME dependence. Configurational and multilevel designs are particularly suitable for identifying combinations of resources, capabilities, institutions, and

ecosystem conditions associated with different BMI outcomes. Finally, future work should specify thresholds and boundary conditions. Research is encouraged to distinguish constraints that stimulate focused experimentation from shortages to identify when networks provide complementary resources or create power asymmetries.

Author contributions

Bi Zhang and Nurul Atasha Jamaludin conducted the literature search, data collection, and bibliometric analysis. Shanshan Yue conceptualized the study, developed the theoretical framework, and led the writing of the manuscript. Xue Lei contributed to the content analysis, interpretation of results, and refinement of the framework. Min Wang provided methodological support and contributed to revising the manuscript for intellectual content. All authors reviewed, edited, and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

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